

potentials: step potential, finite rectangular well, tunnelling from a potential barrier, particle in a box, harmonic oscillator; two and three dimensional systems: concept of degeneracy Quantum states of an electron in an atom; Spectra of one-and many-electron atoms; spin-orbit interactions; fine and hyperfine structures; Zeeman and Stark effects; Electronic, rotational and vibrational spectra of diatomic molecules.

Section 4: Engineering Materials

Classification of Materials, Nature of bonding in Materials, Defects in Crystalline Materials, Mechanical, Structural, Electronic, Thermal, Optical Properties and various Applications of Materials.

Section 5: Synthesis and Characterization of Materials

Top-down and bottom-up synthesis approach, physical and chemical techniques for material synthesis (sol-gel, hydrothermal etc.), X-Ray Diffraction, Scanning Electron Microscopy, Transmission Electron Microscopy, Atomic Force Microscopy, Photoluminescence Spectroscopy and other spectroscopic techniques.

Section 6: Electronic Devices and Circuits

Semiconductors, Metal-semiconductor junctions; Ohmic and rectifying contacts; PN diodes, bipolar junction transistors, field effect devices; negative and positive feedback circuits; oscillators, operational amplifiers, active filters; basics of digital logic circuits, combinational and sequential circuits, flip-flops, timers, counters, registers, A/D and D/A conversion, Opto-electronic devices, Microprocessor and microcontroller basics.

Section 7: Communication systems

Random processes: autocorrelation and power spectral density, properties of white noise, filtering of random signals through LTI systems.

Analog communications: amplitude modulation and demodulation, angle modulation and demodulation, spectra of AM and FM, super heterodyne receivers.

Information theory: entropy, mutual information and channel capacity theorem.

Digital communications: PCM, DPCM, digital modulation schemes (ASK, PSK, FSK, QAM), bandwidth, inter-symbol interference, MAP, ML detection, matched filter receiver, SNR and BER.

Section 8: Electromagnetics

Maxwell's equations: Differential and integral forms and their interpretation, boundary conditions, wave equation, Poynting vector.

Plane waves and properties: reflection and refraction, polarization, Phase and group velocity, propagation through various media, skin depth.

Transmission lines, Rectangular and circular waveguides, light propagation in optical fibers, dipole and monopole antennas, linear antenna arrays.

3. Department of Applied Mathematics

DISCIPLINE: MATHEMATICS

Differential & Integral calculus Vector calculus Algebra (Linear & Abstract) Differential equations (ODE & PDE) Laplace & Fourier transforms, Fourier series. Probability, statistics & operations research Numerical methods, Special functions Real and complex analysis.

DISCIPLINE: MATHEMATICS AND COMPUTING:

Entrance exam will be 30% in Mathematics domain and 70% in Computing Domain

Mathematics Doman (30%)

Calculus: Sequences and Series, differential calculus, integral calculus, vector calculus. Basics and ordinary differential equation (ODE) and partial differential equation (PDE), Fourier Series.

Linear Algebra: Determinants and matrices, Cayley-Hamilton Theorem, Hermitian, skew Hermitian, unitary matrices, eigen values, eigen vectors.

Transforms: Laplace Transform, Fourier Transform.

Numerical Analysis: Numerical solution of algebraic equations using Gauss elimination and Gauss-Siedel methods, Gauss Jordan,

numerical solution of ordinary differential equations using Picard, Euler method. Interpolation.

Probability and Statistics: mean, mode, median and standard deviation, Probability space, conditional probability, Baye's theorem, uniform, binomial, poisson, normal and exponential distribution.

Computing Domain (70%)

Programming: Programming fundamentals & C/C++ programming, object-oriented programming concepts using C++/JAVA.

Data Structures and Algorithms: Arrays, stack, queue, linked list, trees, binary search trees, graph, sorting and searching. Algorithm design techniques: greedy, dynamic programming and divide-and-conquer. Asymptotic space and time complexity.

Digital Logic Design: Boolean Algebra, Logic Functions, combinational and sequential circuit design, registers and counters, logic families.

Computer organization and architecture: Machine instructions and addressing modes, Instruction pipelining, memory hierarchy, I/O devices.

Computer network: OSI, TCP/IP, Brief overview of OSI layers, IPv4, IPv6, routers and routing algorithms (distance vector, link state), congestion control. Application layer protocols (DNS, SMTP, POP, FTP,HTTP).

Database Management System: ER model, relational model, normalization, integrity constraints, SQL, indexing transaction processing, concurrency control.

Theory of Computation: Regular expression, Finite automata, context free grammar, pushdown automata, regular and context free languages, pumping lemma, Turing machine.

Compiler Design: Lexical Analysis, parsing, syntax-directed translation, runtime environments, intermediate code generation.

Operating System: Types of OS, process management, concurrency and synchronization, CPU scheduling, deadlocks, memory management and virtual memory, disk scheduling, file systems.

Discrete Structures: Classical logic theory, Boolean algebra and relations and graph, tree, spanning tree, planar graph and coloring.

Software Engineering: Requirement and Feasibility Analysis, Data flow Diagrams, Process life cycle, design, coding, testing, implementation and maintenance.

4. Department of Civil Engineering

DISCIPLINE: CIVIL ENGINEERING

Section - 1: Civil Engineering (General)

Engineering Surveying, Basic principles of surveying, types and classification of survey, surveying equipment, levelling, Maps and Map making, Indian Map series. Engineering Mechanics, Basic terminology, Units and dimensions, Force, Torque, Friction, Laws of mechanics Free body diagram, solid mechanics. Construction Materials, Types and characteristics of different types of construction materials to include Steel, Concrete, Wood, Stone, Brick/Masonry etc, requirements of good construction materials. Transportation and Highway Engineering – Types and classification of roads, IRC, road construction and materials, road construction equipment. Project Management, Basic terms in project management, , CPM/PERT, critical path, float/slack, quality control, risk management, resource scheduling and management, project management software.

Section - 2 : Mathematics

Linear Algebra - Vector calculus; matrices; similarity transformations, diagonalization, Cayley-Hamilton Theorem, Hermitian, skew Hermitian, unitary matrices, eigenvalues and eigenvectors; linear differential equations: second order linear differential equations; Partial differential equations,

Laplace transform, Fourier series & analysis; Tensors. Numerical Analysis – Numerical Analysis - Numerical solution of algebraic equations using Gauss elimination and Gauss-Siedel methods, numerical solution of ordinary differential equations using Picard, Euler method. Interpolation. Statistics and Probability-mean,mode, median and standard